

SOLUTION

ANSWER. $N = 5$, $M = 16$, and the smallest possible number of elements is

32.

Throughout, a multiset S of positive integers < 20 is encoded by its generating polynomial $f_S(x) = \sum_{s \in S} x^s$, where each element s contributes one monomial x^s counted with multiplicity. Thus f_S has nonnegative integer coefficients, its support lies in the exponents $\{1, 2, \dots, 19\}$, and the number of elements of S (with multiplicity) equals $f_S(1)$.

0. REFORMULATION

LEMMA 1. Let i be a positive integer. S can be partitioned into multisets A_i, B_i with $A_i = i + B_i$ if and only if

$$f_S(x) = (1 + x^i) g(x) \quad \text{for some polynomial } g \text{ with} \\ \text{nonnegative integer coefficients.} \quad (*)$$

Proof. A partition with $A_i = i + B_i$ is the same as a way of grouping the elements of S into unordered pairs $\{b, b+i\}$ (one element b of B_i together with the corresponding element $b+i$ of A_i). If $g(x) = \sum_b (\text{mult}) x^b$ is the generating polynomial of B_i , then A_i has generating polynomial $x^i g(x)$, and $S = A_i \cup B_i$ gives $f_S = (1+x^i)g$. Conversely, given $(*)$, g has nonnegative integer coefficients, so it is the generating polynomial of a multiset B_i , and $A_i := i + B_i$ works.

Hence the index of S is the largest n such that $(*)$ holds for every $i = 1, 2, \dots, n$; if even $i = 1$ fails the index is 0.

In particular, putting $x = 1$ in $(*)$ shows $|S| = f_S(1)$ is even whenever the index is > 1 .

1. A NECESSARY CYCLOTOMIC CONDITION

Let Φ_m denote the m -th cyclotomic polynomial. We use

$$\begin{aligned} \Phi_2 &= 1 + x & \Phi_8 &= 1 + x^4 & \Phi_4 &= 1 + x^2 & \Phi_{10} &= 1 - x + x^2 - x^3 + x^4 & \Phi_6 &= 1 - x + x^2 \\ \Phi_{12} &= 1 - x^2 + x^4 \end{aligned}$$

and the factorizations

$$\begin{aligned} 1 + x &= \Phi_2 & 1 + x^2 &= \Phi_4 & 1 + x^3 &= \Phi_2 \Phi_6 & 1 + x^4 &= \Phi_8 & 1 + x^5 &= \Phi_2 \Phi_{10} \\ 1 + x^6 &= \Phi_4 \Phi_{12} \end{aligned} \quad (1)$$

(These are read off from $1 + x^i = (x^{2i} - 1)/(x^i - 1)$.)

If S has index $> n$ then for every $i > n$, $(1+x^i)$ divides f_S , hence every cyclotomic factor occurring in (1) for $i > n$ divides f_S . Since distinct cyclotomic polynomials are coprime, their product divides f_S . Define

$$L_5 = \Phi_2 \Phi_4 \Phi_6 \Phi_8 \Phi_{10} \quad (\deg 13), \quad L_6 = \Phi_2 \Phi_4 \Phi_6 \Phi_8 \Phi_{10} \Phi_{12} \quad (\deg 17).$$

By (1):

index 5 $\Rightarrow L_5 \mid f_S$, index 6 $\Rightarrow L_6 \mid f_S$. (2)

Their explicit coefficient lists (exponents 0,1,2,...) are

$L_5 = [1, -1, 2, -1, 2, 0, 1, 1, 0, 2, -1, 2, -1, 1]$ $L_6 = [1, -1, 1, 0, 1, 0, 1, 0, 1, 1, 0, 1, 0, 1, 0, 1, -1, 1]$ (3)

(One checks (3) directly by multiplying the cyclotomic polynomials above; note $L_6 = L_5 \cdot \Phi_{12}$.) Both L_5 and L_6 are monic with constant term 1.

2. THE MAXIMUM INDEX IS $N = 5$

2a. Index 6 is impossible.

Suppose S has index 6. By (2), $L_6 \mid f_S$, so $f_S = L_6 \cdot h$ for an integer polynomial h . Because the support of f_S lies in $\{1, \dots, 19\}$ we have $\deg f_S \leq 19$, so $\deg h \leq 19 - 17 = 2$; and $f_S(0) = 0$ with $L_6(0) = 1$ forces $h(0) = 0$. Therefore

$$h(x) = a x + b x^2, \quad a, b \text{ integers.}$$

Using (3), the coefficient of x^2 in f_S is $[x^2] f_S = a \cdot L_6[1] + b \cdot L_6[0] = -a + b$, and the coefficient of x^3 is $[x^3] f_S = a \cdot L_6[2] + b \cdot L_6[1] = a - b$. Both must be 0 (coefficients of a multiset), so $-a+b = 0$ and $a-b = 0$, i.e.

$$a = b =: t.$$

Since S is nonempty, $t \geq 1$, and

$$f_S = t(x + x^2) L_6 = t \cdot x(1+x) L_6.$$

Now test $i = 1$. Dividing by $(1+x) = \Phi_2$ (which is a genuine factor): $f_S / (1+x) = t \cdot x \cdot L_6$. The coefficient of x^2 here equals $t \cdot L_6[1] = -t < 0$. Hence the quotient g in (*) for $i = 1$ has a NEGATIVE coefficient, so there is no valid partition for $i = 1$ — contradicting index 6 (which requires $i = 1$ to work).

Therefore no S has index 6, and $N \leq 5$.

2b. Index 5 is attained.

Take $h(x) = x + 2x^2 + x^3$ and set $f = L_5 \cdot h$. Expanding (using (3)),

$$f(x) = x + x^2 + x^3 + 2x^4 + 2x^5 + 3x^6 + 3x^7 + 3x^8 + 3x^9 + 3x^{10} + 3x^{11} + 2x^{12} + 2x^{13} + x^{14} + x^{15} + x^{16},$$

which corresponds to the multiset

$$S^* = \{1, 2, 3, 4, 4, 5, 5, 6, 6, 6, 7, 7, 7, 8, 8, 8, 9, 9, 9, 10, 10, 10, 11, 11, 11, 12, 12, 13, 13, 14, 15, 16\}.$$

All coefficients are nonnegative and the support is $\{1, \dots, 16\} \subseteq \{1, \dots, 19\}$, so S^* is a legitimate multiset. Its index is exactly 5: explicit partitions into pairs $\{b, b+i\}$ for $i = 1, 2, 3, 4, 5$ (each a list of 16 pairs covering S^* exactly) are

$i=1$: (1,2)(3,4)(4,5)(5,6)(6,7)(6,7)(7,8)(8,9)(8,9)(9,10)(10,11)(10,11)(11,12)(12,13)(13,14)(15,16)
 $i=2$: (1,3)(2,4)(4,6)(5,7)(5,7)(6,8)(6,8)(7,9)(8,10)(9,11)(9,11)(10,12)(10,12)(11,13)(13,15)(14,16)
 $i=3$: (1,4)(2,5)(3,6)(4,7)(5,8)(6,9)(6,9)(7,10)(7,10)(8,11)(8,11)(9,12)(10,13)(11,14)(12,15)(13,16)
 $i=4$: (1,5)(2,6)(3,7)(4,8)(4,8)(5,9)(6,10)(6,10)(7,11)(7,11)(8,12)(9,13)(9,13)(10,14)(11,15)(12,16)
 $i=5$: (1,6)(2,7)(3,8)(4,9)(4,9)(5,10)(5,10)(6,11)(6,11)(7,12)(7,12)(8,13)(8,13)(9,14)(10,15)(11,16)

(One verifies for each i that the multiset of all entries appearing is exactly S^* .) Thus partitions exist for $i = 1, \dots, 5$, so the index is ≤ 5 ; by part 2a it cannot exceed 5. Hence the index of S^* equals 5, and combined with 2a,

$$N = 5.$$

[Sufficiency remark used implicitly: whenever $f = L_5 \cdot h$ with h having nonnegative coefficients, dropping any single factor $(1+x^i)$ for $i \leq 5$ leaves a product of (3) with nonnegative pieces and h , but in general we simply verify each quotient directly, as done above.]

3. THE SMALLEST MAXIMUM ELEMENT IS $M = 16$

Let S have index 5 ($= N$). By (2), $L_5 \mid f_S$, so $f_S = L_5 \cdot h$ with h an integer polynomial, $h(0) = 0$ (since $f_S(0)=0$, $L_5(0)=1$). The maximum element of S equals $\deg f_S = 13 + \deg h$. Since $h(0) = 0$ and $h \neq 0$, $\deg h \geq 1$, giving maximum element ≥ 14 . We rule out 14 and 15.

- Maximum element 14 ($\deg h = 1$): $h = c \cdot x$, $c \geq 1$, so $f_S = c \cdot x \cdot L_5$. By (3), $[x^2] f_S = c \cdot L_5[1] = -c < 0$, impossible for a multiset.
- Maximum element 15 ($\deg h = 2$): $h = c_1 x + c_2 x^2$ with $c_2 \geq 1$. Then $[x^2] f_S = c_1 \cdot L_5[1] + c_2 \cdot L_5[0] = -c_1 + c_2 \geq 0 \Rightarrow c_2 \geq c_1$, $[x^{14}] f_S = c_1 \cdot L_5[13] + c_2 \cdot L_5[12] = c_1 - c_2 \geq 0 \Rightarrow c_1 \leq c_2$, so $c_1 = c_2 =: s \geq 1$ and $f_S = s \cdot x(1+x)L_5$. Dividing by $(1+x)$, $f_S/(1+x) = s \cdot x \cdot L_5$, whose $[x^2]$ -coefficient is $s \cdot L_5[1] = -s < 0$, so the $i = 1$ partition fails — contradicting index 5.

Hence every index-5 multiset has maximum element ≥ 16 , and S^* above achieves 16. Therefore $M = 16$.

4. THE SMALLEST NUMBER OF ELEMENTS IS 32

Let S have index 5 and maximum element 16. As in Section 3, $f_S = L_5 \cdot h$ with $h(0) = 0$ and $\deg f_S = 16$, so $\deg h = 3$:

$$h(x) = c_1 x + c_2 x^2 + c_3 x^3, \quad c_3 \geq 0.$$

KEY IDENTITY (size formula). Evaluating at $x = 1$, $|S| = f_S(1) = L_5(1) \cdot h(1) = 8 \cdot (c_1 + c_2 + c_3)$, because $L_5(1) = \text{Phi}_2(1)\text{Phi}_4(1)\text{Phi}_6(1)\text{Phi}_8(1)\text{Phi}_{10}(1) = 2 \cdot 2 \cdot 1 \cdot 2 \cdot 1 = 8$. Thus $|S|$ is 8 times the integer $h(1) = c_1 + c_2 + c_3$, and minimizing $|S|$ means minimizing $c_1 + c_2 + c_3$.

Sign constraints on the coefficients of $f_S = L_5 \cdot h$ (read off via (3)): $[x^1] f_S = c_1 \cdot L_5[0] = c_1 \geq 0$, $[x^{16}] f_S = c_3 \cdot L_5[13] = c_3 \geq 0$, and $c_3 \geq 0$, $[x^2] f_S = c_1 \cdot L_5[1] + c_2 \cdot L_5[0] = c_2 - c_1 \geq 0$, $[x^{15}] f_S = c_2 \cdot L_5[13] + c_3 \cdot L_5[12] = c_2 - c_3 \geq 0$. Therefore $c_1 \geq 0$, $c_3 \geq 1$, $c_2 \geq c_1$, $c_2 \geq c_3 \geq 1$. In particular all of c_1, c_2, c_3 are nonnegative with $c_2 \geq 1$ and $c_3 \geq 1$.

We claim $c_1 + c_2 + c_3 \geq 4$. Suppose instead $c_1 + c_2 + c_3 \leq 3$. Subject to $0 \leq c_1 \leq c_2$, $1 \leq c_3 \leq c_2$, the only integer triples are

$$(c_1, c_2, c_3) = (0, 1, 1), (1, 1, 1), (0, 2, 1).$$

We check each fails to give an index-5 multiset:

(0,1,1): $f = L_5(x^2+x^3) = x^2(1+x)L_5$. Its $(1+x)$ -quotient is $x^2 L_5$, whose $[x^3]$ -coefficient is $L_5[1] = -1 < 0$, so $i = 1$ already fails (index 0).

(1,1,1): $f = x(1+x+x^2)L_5$. Direct division by $(1+x)$ gives a quotient with a negative coefficient (e.g. $[x^2]$ of $f/(1+x)$ is -1), so $i = 1$ fails (index 0).

(0,2,1): $f = L_5(2x^2 + x^3)$ has $[x^3] f = 2 \cdot L_5[1] + L_5[0] = -2 + 1 = -1 < 0$, so f is not even a valid multiset.

(Each statement is a one-line coefficient computation from (3).) None of the three has index 5, so $c_1 + c_2 + c_3 \leq 4$, hence

$$|S| = 8(c_1+c_2+c_3) \leq 8 \cdot 4 = 32.$$

This bound is attained: S^* from Section 2b has $h = x + 2x^2 + x^3$, so $c_1+c_2+c_3 = 4$ and $|S^*| = 32$, with index 5 and maximum element 16.

5. CONCLUSION

The maximum index over all admissible multisets is $N = 5$. Among multisets of index 5, the smallest possible maximum element is $M = 16$. Among multisets of index 5 with maximum element 16, the number of elements is a multiple of 8 and is at least 32, with equality achieved by

$$S^* = \{1, 2, 3, 4^2, 5^2, 6^3, 7^3, 8^3, 9^3, 10^3, 11^3, 12^2, 13^2, 14, 15, 16\}.$$

Hence the smallest possible number of elements of S (with multiplicity) is